

A/B Test Analysis of Email Subject Lines for H&M Summer Deal Days

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Introduction

H&M's Marketing Team conducted an A/B test to compare the effectiveness of two different email subject lines on customer engagement during the Summer Deal Days campaign. The control group received emails with the subject line **"20% off + free shipping"**, while the test group received emails with the subject line **"Your last chance to score 20% off + free shipping"**. The primary objective was to determine which subject line generates a higher email open rate to boost engagement and drive sales.

Hypotheses

- **Null Hypothesis (H_0):** There is no difference in email open rates between the control subject line and the test subject line.
- **Alternative Hypothesis (H_1):** The test subject line leads to a higher email open rate than the control subject line.

Data Summary

A total of 9,437 emails were sent, split almost evenly between the control group (4,764 emails) and the test group (4,673 emails). The emails opened were 836 for the control group and 894 for the test group.

Open Rate Analysis

The open rate was calculated for each group:

- Control group open rate: $(836 \div 4,764) \times 100 = 17.55\%$
- Test group open rate: $(894 \div 4,673) \times 100 = 19.13\%$

The test group outperformed the control by approximately **1.58 percentage points**.

Statistical Significance and Interpretation

The increase in open rate for the test group was found to be statistically significant with 98% confidence, meaning there is strong evidence that the subject line used in the test group leads to better engagement. Therefore, the alternative hypothesis—that the new subject line improves open rates—is supported.

Confounding Factors and Mitigation

Two potential confounding factors that could affect the results include:

- **Send Time Differences:** If emails were sent at different times, this might impact open rates. To mitigate this, emails should be sent simultaneously across all segments and time zones.

- **Email List Quality / Randomization:** Uneven randomization could cause one group to have more engaged subscribers. Proper random assignment based on past engagement metrics is essential to balance the groups.

Balancing Open Rates and Conversion

While the new subject line increased open rates, it is critical to ensure this translates into meaningful sales. Tracking additional metrics such as click-through rates, purchase conversions, average order value, and customer lifetime value is necessary to assess overall campaign success. If open rates rise but purchases do not, further optimization of email content or follow-up strategies may be needed.

Recommendation

Given the statistically significant improvement in open rates, it is recommended that H&M adopt the test subject line **“Your last chance to score 20% off + free shipping”** for future Summer Deal Days promotional emails. Monitoring downstream engagement and sales metrics should continue to ensure the campaign’s effectiveness.

References

- Statsig Sample Size Calculator, retrieved from: <https://www.statsig.com/calculator>
- Neil Patel A/B Testing Calculator, retrieved from: <https://neilpatel.com/ab-testing-calculator/>